Running YOLO Object Detection on NVIDIA Jetson Orin Nano

CS 4391 Introduction Computer Vision

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JetPack 6.2 image

- https://developer.nvidia.com/embedded/jetpack-sdk-62

- JetPack 6.2 is the latest production release of JetPack 6. This release includes Jetson Linux 36.4.3, featuring the Linux Kernel 5.15 and an Ubuntu 22.04-based root file system. The Jetson AI stack packaged with JetPack 6.2 includes CUDA 12.6, TensorRT 10.3, cuDNN 9.3, VPI 3.2, DLA 3.1, and DLFW 24.0.

Docker

- An open platform that enables you to separate your applications from your infrastructure

- Container
  - A lightweight environment that contains everything to run an application
  - A container is a runnable instance of an image

- Image
  - A read-only template with instructions for creating a docker container

Step 1: Download the Ultralytics YOLOv8 Docker Image

- https://www.jetson-ai-lab.com/archive/tutorial_ultralytics.html

- t=ultralytics/ultralytics:latest-jetson-jetpack6
- sudo docker pull $t && sudo docker run -it --ipc=host --runtime=nvidia $t

- Then exit the docker environment

Step 2: Start the Docker image

- xhost +local:

- sudo docker run -it --ipc=host --runtime=nvidia --network host --device=/dev/video0:/dev/video0 -e DISPLAY=$DISPLAY -e QT_X11_NO_MITSHM=1 -v /tmp/.X11-unix:/tmp/.X11-unix:rw -v /home/yuxiang/Documents/cs4391_spring26:/ultralytics/cs4391_spring26 -v /tmp/argus_socket:/tmp/argus_socket ultralytics/ultralytics:latest-jetson-jetpack6

Step 3: Install non-headless OpenCV in the Docker

- pip uninstall -y opencv-python-headless
- pip install opencv-python==4.11.0.86

- apt-get update && apt-get install -y \
  libgtk2.0-dev \
  pkg-config \
  libgl1-mesa-glx \
  libglib2.0-0

- https://chatgpt.com/share/69b077af-62b8-800a-b9d3-590056733a42

Step 4: Test your USB camera in Docker

- Using this github repo https://github.com/yuxng/cs4391_spring26.git

- python test_usb_camera.py

Step 5: Commit your Docker Image for reuse

- Exit the docker environment
- Sudo docker container list –a
  - Find the container ID
- Sudo docker container commit <Container_ID>
- Sudo docker image list –a
- Sudo docker image tag <IMGA_ID> TAG

Step 5: Test your USB camera in the new Docker Image

- Run the same docker command
- Using this github repo https://github.com/yuxng/cs4391_spring26.git
- python test_usb_camera.py
- It should work without installing the non-headless opencv now

Step 6: Test YOLO on offline images

- Using this github repo https://github.com/yuxng/cs4391_spring26.git
- python test_YOLO.py
- The code will build the TensorRT for yolo, which may take some time

Step 7: Test YOLO with the USB camera

- python test_YOLO_usb_camera_docker.py

Questions?